

Sample Report: Random Forest vs. Other Binary Classifiers on the Ionosphere Dataset

Sample Student
COMP 395 – Deep Learning

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Task and methods

The UCI Ionosphere dataset contains 351 radar returns, each described by 34 numeric features and labeled *good* (the signal shows structure in the ionosphere) or *bad*. It is close in size and difficulty to the breast cancer data from Lab 2. I used a stratified 80/20 split (280 training and 71 test samples) and normalized every feature with the training mean and standard deviation. I compared a random forest (200 trees) against five other methods: the single sigmoid neuron from Lab 2 (MSE loss, SGD, $\alpha = 0.01$, 100 epochs), logistic regression, k -nearest neighbors ($k = 5$), an RBF-kernel SVM, and a single decision tree. All scikit-learn models used default settings.

Results

Table 1 and Figure 1 show that the random forest and the SVM tied for the best test accuracy (95.8%), followed by my from-scratch neuron (94.4%). Figure 2 shows the neuron’s loss falling quickly for about ten epochs and then flattening, so more epochs would help very little.

Table 1: Accuracy of each method. CV is the mean of 5-fold cross-validation on the training set (not run for the from-scratch neuron).

Method	Train	Test	5-fold CV
Sigmoid neuron (from scratch)	0.925	0.944	–
Logistic regression	0.932	0.930	0.879
k -nearest neighbors ($k = 5$)	0.868	0.887	0.811
SVM (RBF kernel)	0.964	0.958	0.936
Decision tree	1.000	0.901	0.875
Random forest (200 trees)	1.000	0.958	0.914

Discussion

The two best models are the two that can draw curved decision boundaries, which suggests the classes are not linearly separable; the linear models (my neuron and logistic regression) land a few points lower. The single decision tree memorized the training set (100% train, 90.1% test), a clear sign of overfitting. The random forest also reaches 100% on the training set, but averaging 200

trees, each grown on a different bootstrap sample, cancels much of that variance and recovers about six points of test accuracy. k -NN did worst, probably because distances become less informative across 34 dimensions. One caution: with only 71 test samples, each sample is worth 1.4 percentage points, so the gap between the top three methods is one or two radar returns. The cross-validation column is the steadier comparison, and there the SVM leads the random forest by about two points.

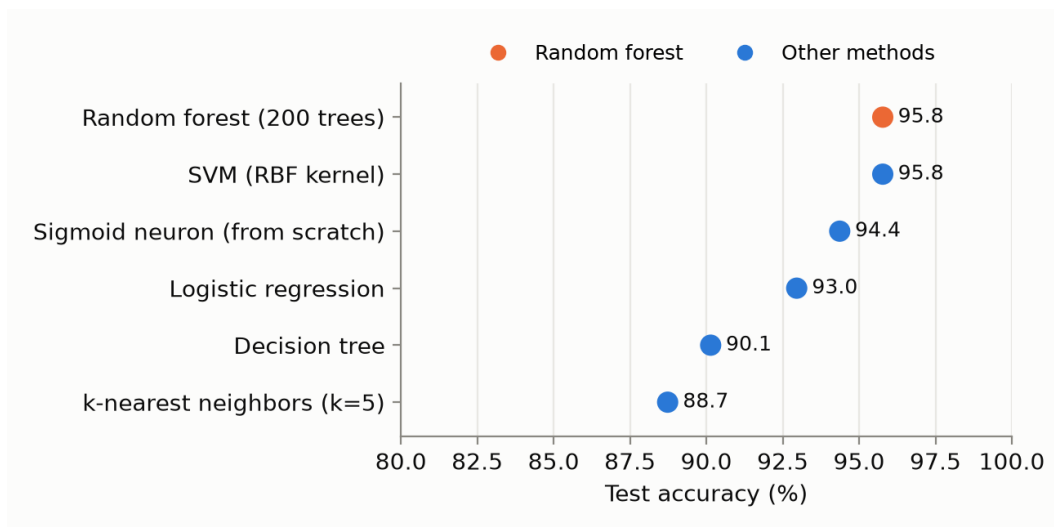


Figure 1: Test accuracy of each method (71 test samples).

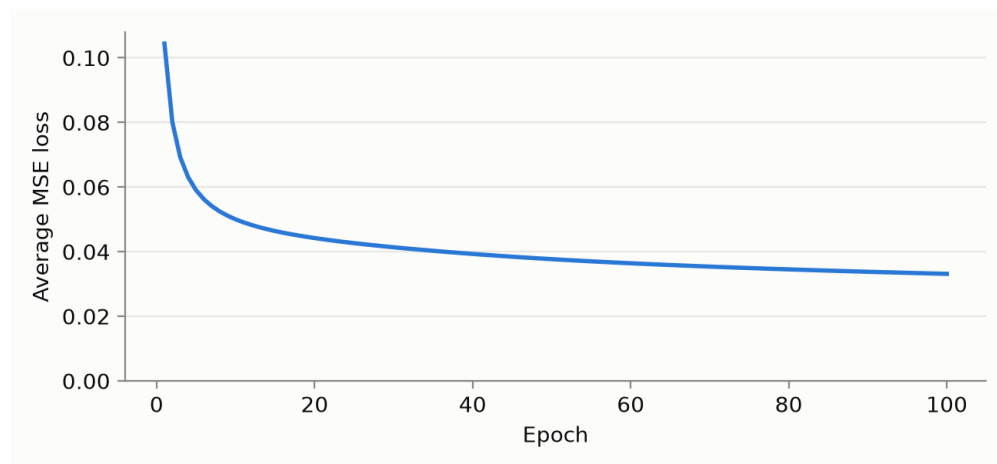


Figure 2: Average training loss per epoch for the from-scratch sigmoid neuron.